Lecture 3

CSS3223 Calculus

CHAPTER 2 :

LIMITS AND CONTINUITY

2.1 RATES OF CHANGE AND TANGENTS TO CURVES

2.2 LIMIT OF A FUNCTION AND LIMIT LAWS

2.3 ONE-SIDED LIMITS AND LIMITS AT INFINITY

2.4 INFINITE LIMITS AND VERTICAL ASYMPTOTES

2.5 CONTINUITY

2.6 TANGENTS AND DERIVATIVES AT A POINT

Rates of Change and Tangents to Curves

- Objectives:
  - Understand what average is
  - Understand what instantaneous rates of change is
  - Relating the above with slope of a curve at a point on the curve
  - Introduce limit
  - Using limit to calculate the slope

Average Speed

- Average speed of a moving body is

$$\frac{\text{distance covered}}{\text{time elapsed}}$$

Can you guess what is the unit for average speed?

Average and Instantaneous Speed

Example 1

A rock breaks loose from the top of a tall cliff forming a distance function

$$y = 16t^2 \quad \text{feet}$$

What is its average speed during

a) The first 2 sec fall?

b) The 1-sec interval between second 1 and second 2?

Average and Instantaneous Speed

Continuation of example 1,

Find the speed of the falling rock at t =1 and t =2.

TABLE 2.1 Average speeds over short time intervals

$$\text{Average speed: } \frac{\Delta y}{\Delta t} = \frac{16(t_0 + h)^2 - 16{t_0}^2}{h}$$

Length of time interval h	Average speed over interval of length h starting at $t_0 = 1$	Average speed over interval of length h starting at $t_0 = 2$
1	48	80
0.1	33.6	65.6
0.01	32.16	64.16
0.001	32.016	64.016
0.0001	32.0016	64.0016

Average and Instantaneous Speed

Let's look at this equation.

$$\frac{\Delta y}{\Delta t} = \frac{16(1+h)^2 - 16(1)^2}{h}$$

If we set t_0=1, and then expand the numerator in the equation and simplify, we get that

$$\frac{\Delta y}{\Delta t} = 32 + 16h$$

As *h* approaches 0, the average speed has a limiting value 32+16(0) = 32 ft/sec.

Average Rates of Change

DEFINITION The **average rate of change** of $y = f(x)$ with respect to x over the interval $[x_1, x_2]$ is

$$\frac{\Delta y}{\Delta x} = \frac{f(x_2) - f(x_1)}{x_2 - x_1} = \frac{f(x_1 + h) - f(x_1)}{h}, \qquad h \neq 0.$$

To determine **Instantaneous Rate of Change (speed),** we take consideration of this equation when h approaches 0.

$$\lim_{h \to 0} \frac{f(x_1 + h) - f(x_1)}{h}$$

Average rates of change and Secant lines

- Geometrically, the rate of change of f over $[x_1, x_2]$ is the slope of the line through the points $P(x_1,f(x_1))$. and $Q(x_2,f(x_2))$.

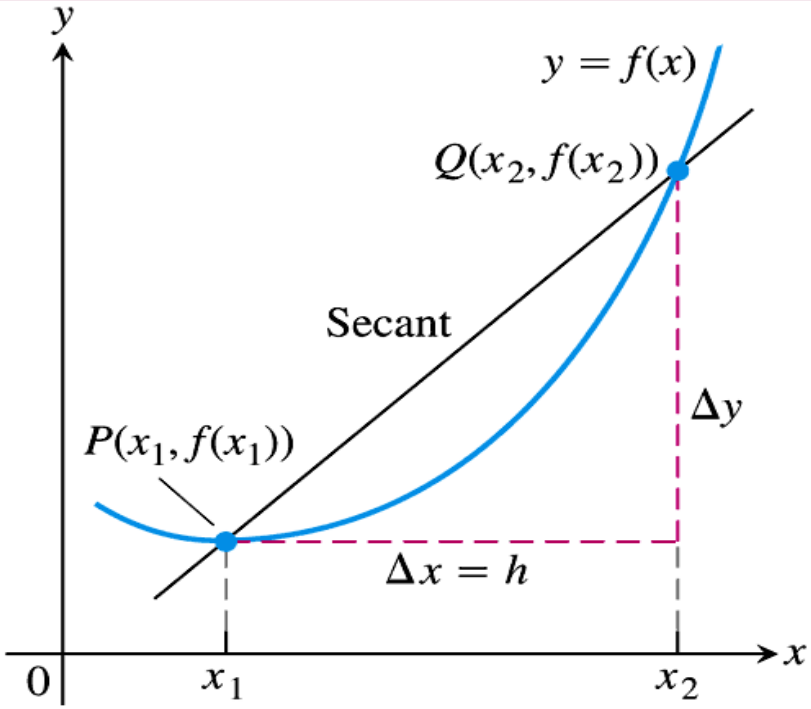


FIGURE 2.1 A secant to the graph $y = f(x)$. Its slope is $\Delta y/\Delta x$, the average rate of change of f over the interval $[x_1, x_2]$.

Tangent of a circle at a point P

What is a tangent line ?

A line L is tangent to a circle at a point P if L passes through P perpendicular to the radius at P

A line joining two points of a curve is a secant to the curve.

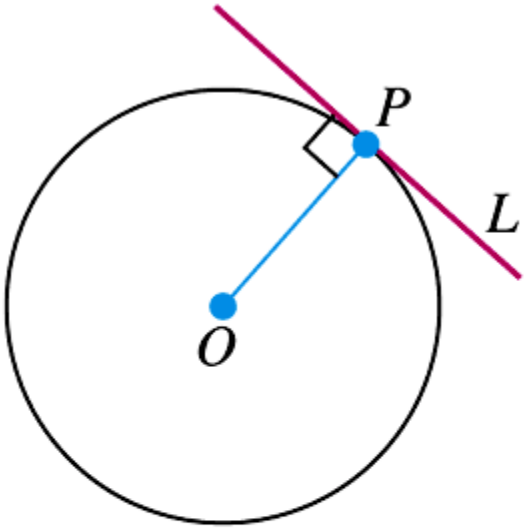


FIGURE 2.2 L is tangent to the circle at P if it passes through P perpendicular to radius OP.

Tangent to some other curve at a point P

- To define the above:
  - Consider two points on the curve, P and Q
  - Join the two points – this is called the **secant** to the curve
  - Calculate the slope of the secant PQ
  - Investigate the behaviour of the slope as Q approaches P
  - If there still a value for the slope at P then we say the tangent to the curve at P is the line through P with the slope.

Tangent of a curve at a point

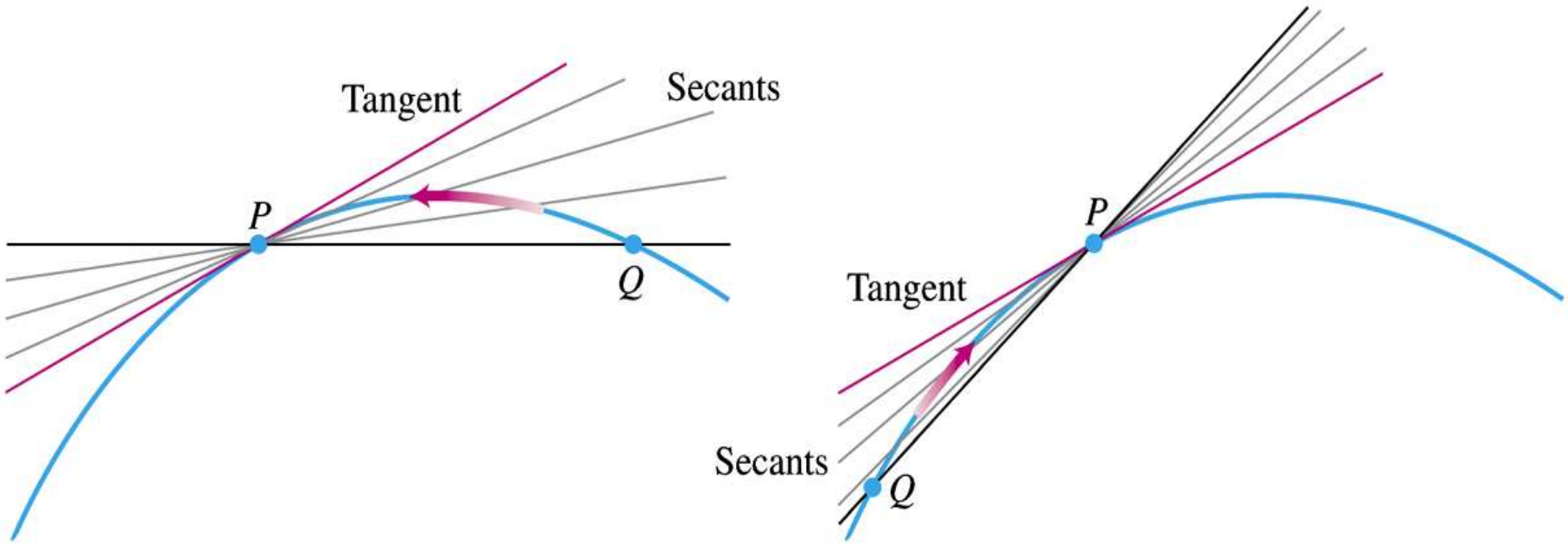
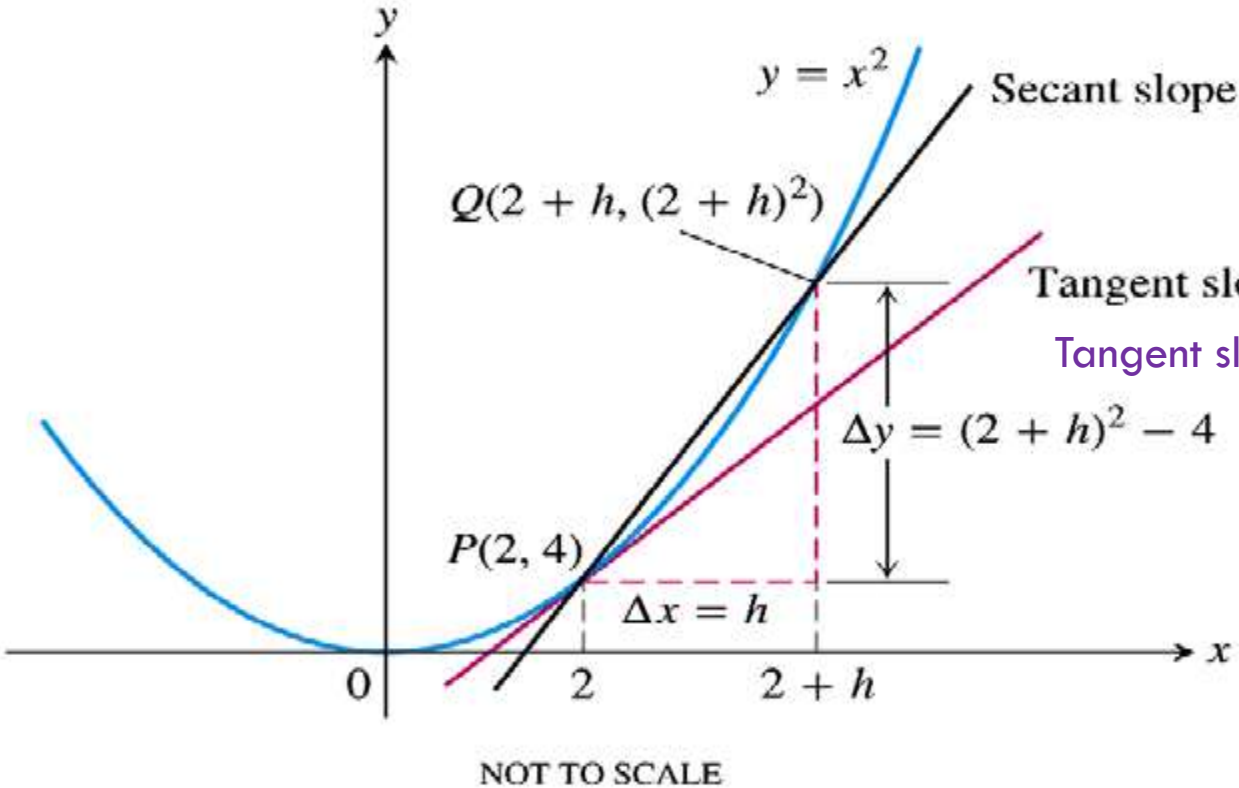


FIGURE 2.3 The tangent to the curve at P is the line through P whose slope is the limit of the secant slopes as $Q \to P$ from either side.



Tangent slope $= \lim_{h \to 0} h + 4 = 0 + 4 = 4$

FIGURE 2.4 Finding the slope of the parabola $y = x^2$ at the point $P(2, 4)$ as the limit of secant slopes.

Example

- Find the slope of the parabola $y = x^2$ at the point $P(2,4)$. Write an equation for the tangent to the parabola at this point.
- Tangent slope $= \lim_{h \to 0} h + 4 = 0 + 4 = 4$ or

$$\frac{dy}{dx} = 2x,$$

gradient of the tangent, $m = 2x = 2(2) = 4$,

equation of the tangent :

$$y - y_1 = m(x - x_1)$$

$$y - 4 = 4(x - 2)$$

$$y = 4x - 8 + 4 = 4x - 4 = 4(x - 1).$$

Instantaneous Rates of Change

Instantaneous rates and slopes of tangent lines are intimately connected.

Example:

A population of fruit flies that grew in a 50-day experiment was done. The number of flies was counted at regular intervals. The counted values are plotted with respect to time, and the points are joined by a smooth curve. Find the average growth rate from day 23 to day 45

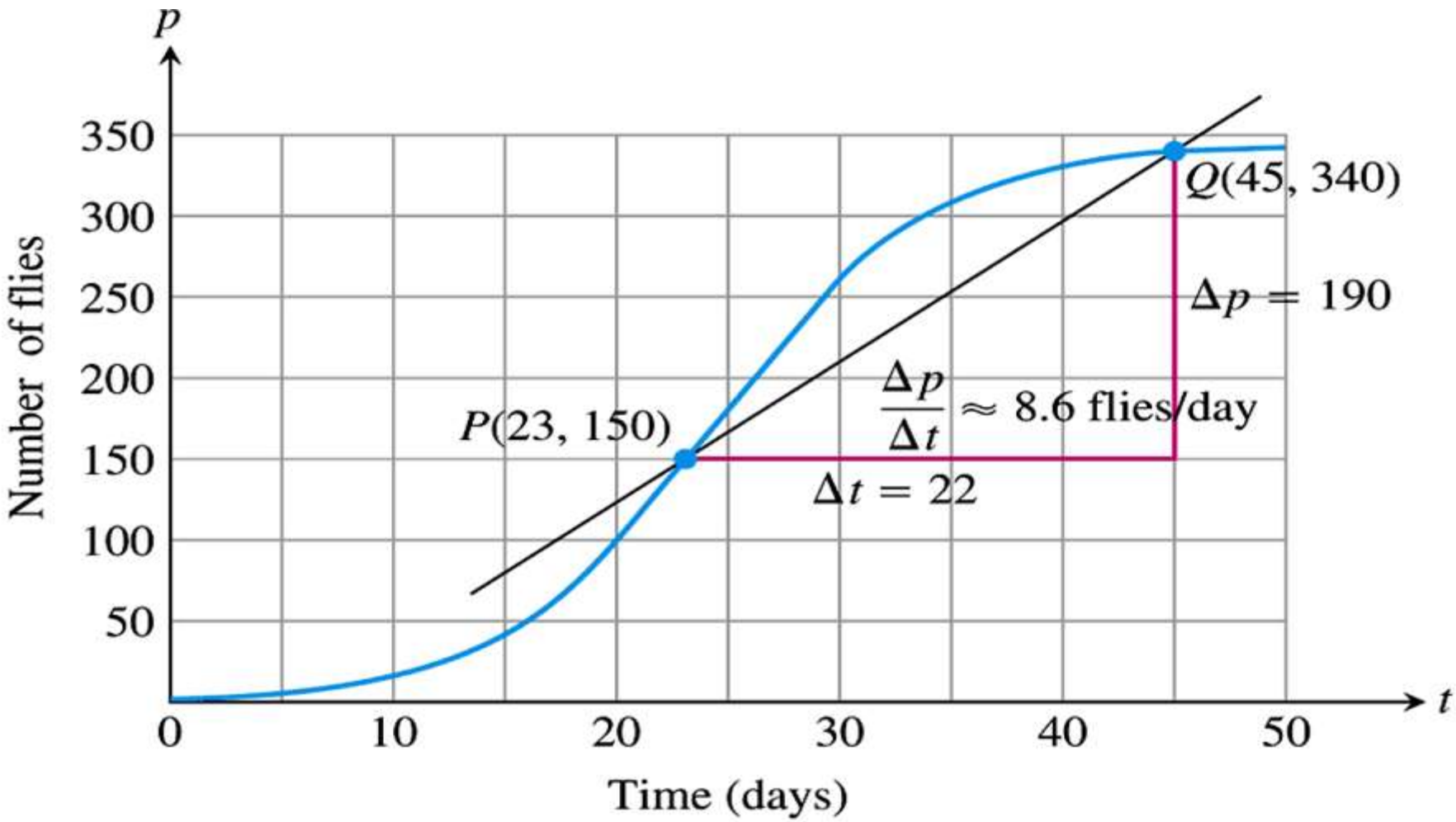


FIGURE 2.5 Growth of a fruit fly population in a controlled experiment. The average rate of change over 22 days is the slope $\Delta p/\Delta t$ of the secant line.

Note on the previous example

- The average rate of change from day 23 to day 45 does not tell us how fast the population was changing on day 23 itself.
- To calculate this, we need to examine time intervals closer to the day.

Continuation to the Example

- How fast was the number of flies in the population growing?
- To answer this, we examine the average rates of change over increasingly short intervals starting at day 23
- i.e we calculate the slopes of secant from P to Q, for a sequence of points Q approaching P along the curve.

Q	Slope of $PQ = \Delta p/\Delta t$ (flies/day)
(45, 340)	$\frac{340 - 150}{45 - 23} \approx 8.6$
(40, 330)	$\frac{330 - 150}{40 - 23} \approx 10.6$
(35, 310)	$\frac{310 - 150}{35 - 23} \approx 13.3$
(30, 265)	$\frac{265 - 150}{30 - 23} \approx 16.4$

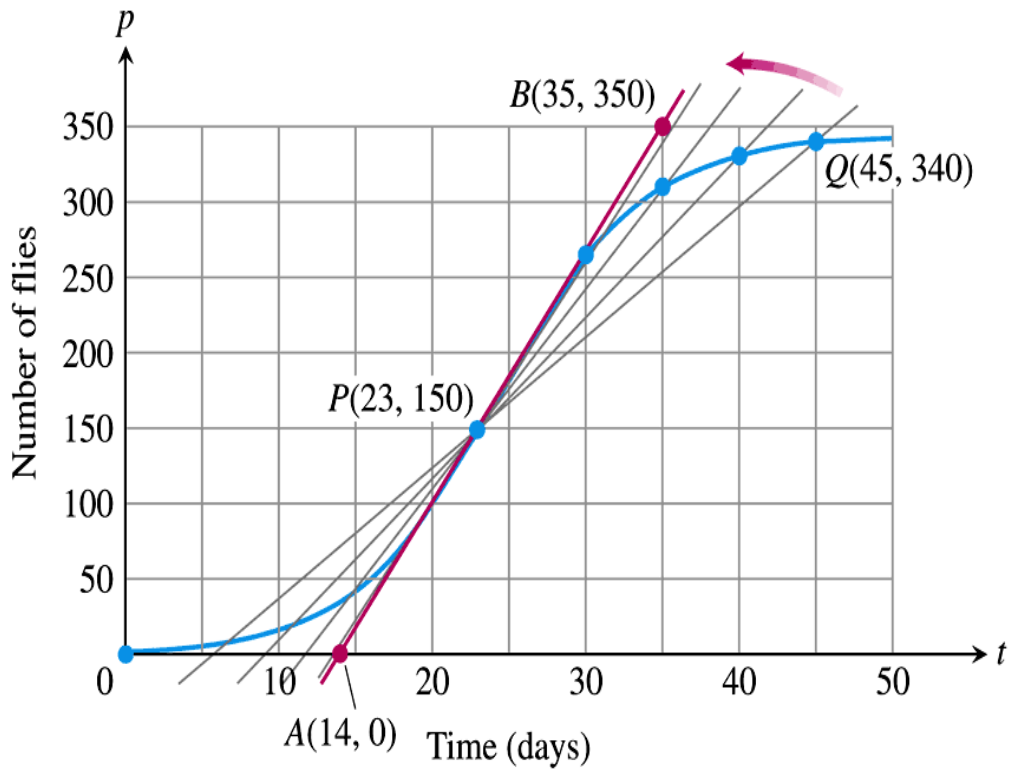


FIGURE 2.6 The positions and slopes of four secants through the point P on the fruit fly graph (Example 5).

The Difference Quotient and the Derivative

Difference Quotient	Derivative
$\dfrac{f(x+h)-f(x)}{h}$	$\lim\limits_{h \to 0} \dfrac{f(x+h)-f(x)}{h}$
■ Slope of the secant line	■ Slope of the tangent line
■ Average rate of change	■ Instantaneous rate of change
■ Average velocity	■ Instantaneous velocity
■ Average rate of change in cost, revenue, or profit	■ Marginal cost, revenue, or profit

Learning outcome

- You got the “limit” idea
- Recall:
  - Instantaneous rates = limiting values of average rates
    = the slope of the tangent line to a curve at a point
    = limiting values of the slopes of the secant lines

Limit of a function and Limit Laws

- Objectives:
  - To understand the informal/intuitive definition of limit
  - To calculate the values of limits

What is a limit?

- Suppose we want to calculate the area of a square.
- One way to do this is to calculate the length of a side and square it to get the area.

- Example: if the side is 4 then the area is

  $4^2 = 16$
- Suppose we change the problem slightly; if the length of the side is close to 4, then is the area close to 16?

What is a limit?

- If the length of the side of a square is close to 4 then is the area close to 16?
- From table, it appears that if the side is close to 4, then the area is close to 16

side	area
4.1	16.81
4.01	16.0801
4.001	16.008001
4.0001	16.00080001
4.00001	16.0000800001

Note also as the side gets closer and closer to 4, the area gets closer and closer to 16
And to get the area close to 16, we can let the side as close to 4

Thus, the limit of the area as the side approaches 4 is 16

What is a limit?

- Using notation:
- Let x denote the side and A(x) is the area
- Then, if x close to 4 then A(x) is close to 16
- Also, as x gets closer and closer to 4, then A(x) gets closer and closer to 16
- And, we can get A(x) close to 16 by making x get closer to 4
- Thus, the limit of A(x) as x approaches 4 is 16 where $A(x) = x^2$

What is a limit?

- Another example:

  $g(x) = 3x - 2$
- $g(7) = 19$; if x is close to 7, is g(x) close to 19?

x	g(x)
7.1	19.3
7.01	19.03
7.001	19.003
7.0001	19.0003
7.00001	19.00003

Same conclusion as before

What is a limit?

- In each of these two examples, we have a function f and two numbers a and L
  - $f(x) = A(x) = x^2$, a = 4 and L = 16
  - $f(x) = g(x) = 3x - 2$, a = 7 and L = 19
- Both examples have three conditions:
  1. If x is close to a, then f(x) is close to L
  2. As x gets closer and closer to a, then f(x) gets closer and closer to L
  3. We can make f(x) as close to L as we want by making x close to a

First intuitive definition of a limit

If the three conditions hold then the limit of f(x) as x approaches a is L

1. If x is close to a, then $f(x)$ is close to L
2. As x gets closer and closer to a, then $f(x)$ gets closer and closer to L
3. We can make $f(x)$ as close to L as we want by making x close to a

- Now consider:

$$f(x) = \frac{x^2 - 1}{x - 1}$$

- What happens as x approaches 1?

x	f(x)
1.1	2.1
1.01	2.01
1.001	2.001
1.0001	2.0001
1.00001	2.00001

It seems that the limit of f(x) as x approaches 1 is 2

Need to make sure that $x \neq 1$ because f(1) is undefined

Require, as x gets closer and closer to 1, x is not equal to 1

Intuitive definition of a limit

- If f is a function and a and L are numbers such that

1. If x is close to a but not equal to a then $f(x)$ is close to L
2. As x gets closer and closer to a but not equal to a then $f(x)$ gets closer and closer to L
3. We can make $f(x)$ as close to L by making x close to a but not equal to a
   - then the limit of $f(x)$ as x approaches a is L

$$\lim_{x \to a} f(x) = L$$

Using the notation, our examples:

- f(x) = A(x) = x^2, a = 4 and L = 16
- Then, $\lim_{x \to 4} A(x) = \lim_{x \to 4} x^2 = 16$
- f(x) = g(x) = 3x-2, a = 7 and L = 19
- Then, $\lim_{x \to 7} g(x) = \lim_{x \to 7} 3x - 2 = 19$
- $f(x) = \dfrac{x^2 - 1}{x - 1}$, a = 1 and L = 2
- Then, $\lim_{x \to 1} f(x) = \lim_{x \to 1} \dfrac{x^2 - 1}{x - 1} = 2$

Recollection

- We have seen:
  - Limits arise when finding the instantaneous rate of change of a function or the tangent to a curve
  - Informal definition of limit

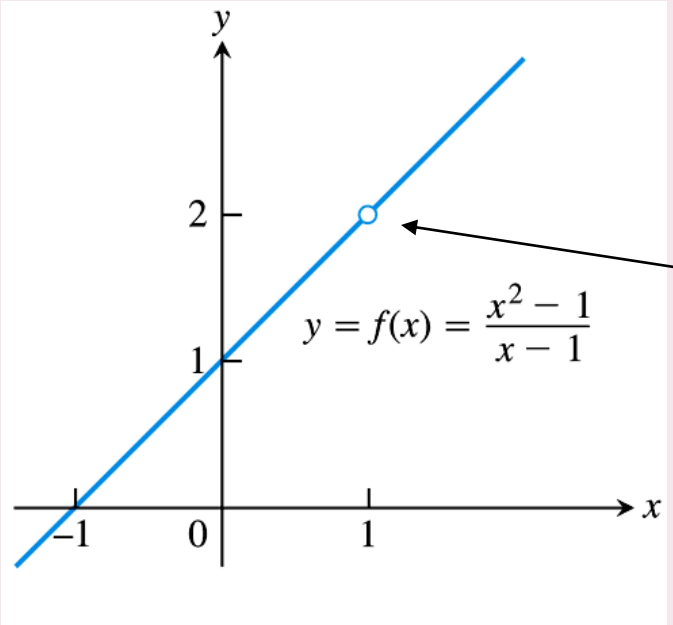
Limit of a function and Limit Laws

- Objectives:
  - Calculate the values of limits
  - Learn some limit laws

Limits of Function Values

Example

How does the function $f(x) = \dfrac{x^2 - 1}{x - 1}$ behave near $x = 1$



This point is removed creating a hole

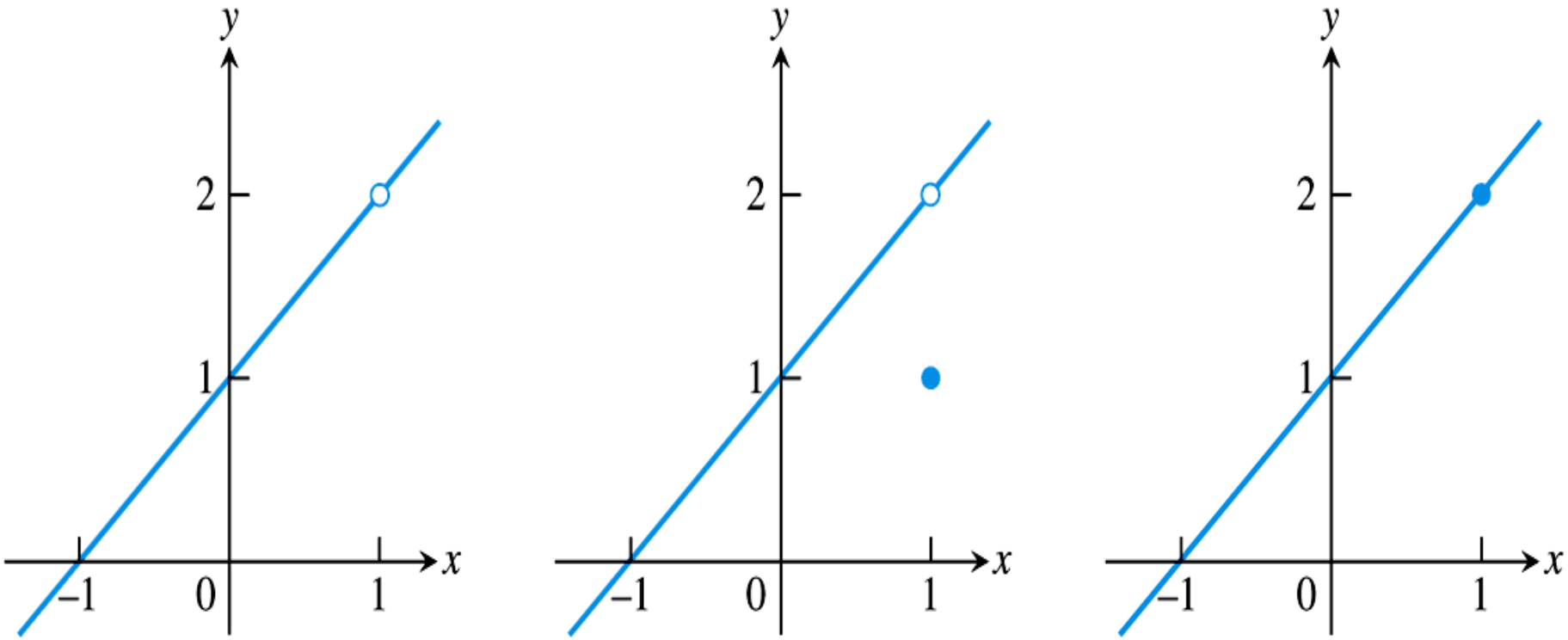
But, what happens when *x nears* to 1?

TABLE 2.2 The closer x gets to 1, the closer $f(x) = (x^2 - 1)/(x - 1)$ seems to get to 2

Values of x below and above 1	$f(x) = \dfrac{x^2 - 1}{x - 1} = x + 1, \quad x \neq 1$
0.9	1.9
1.1	2.1
0.99	1.99
1.01	2.01
0.999	1.999
1.001	2.001
0.999999	1.999999
1.000001	2.000001

We say that f(x) approaches the limit 2 as x approaches 1, and write

$$\lim_{x \to 1} \frac{x^2 - 1}{x - 1} = 2$$



(a) $f(x) = \dfrac{x^2 - 1}{x - 1}$

(b) $g(x) = \begin{cases} \dfrac{x^2 - 1}{x - 1}, & x \neq 1 \\ 1, & x = 1 \end{cases}$

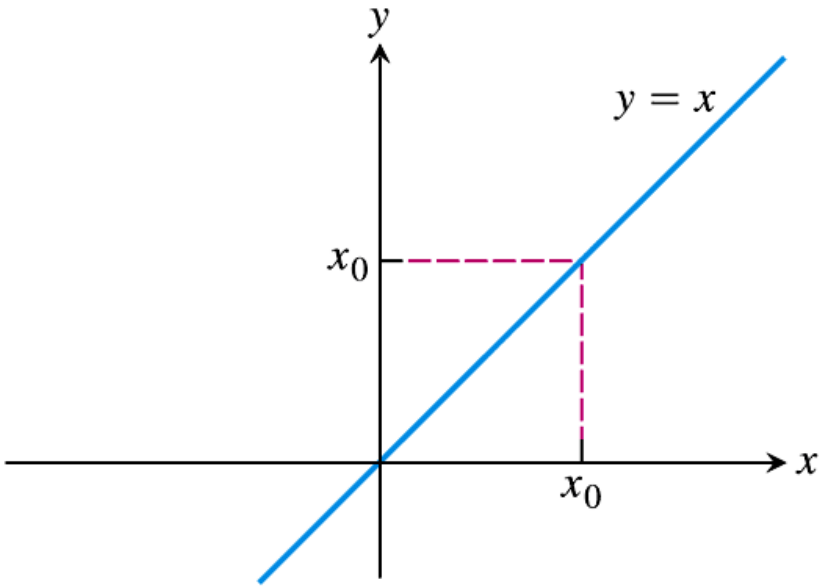
(c) $h(x) = x + 1$

FIGURE 2.8 The limits of $f(x)$, $g(x)$, and $h(x)$ all equal 2 as x approaches 1. However, only $h(x)$ has the same function value as its limit at $x = 1$ (Example 2).

Identity and Constant Functions

- If f is the **identity function** $f(x) = x$, then for any value of x_0

$$\lim_{x \to x_0} f(x) = \lim_{x \to x_0} x = x_0$$

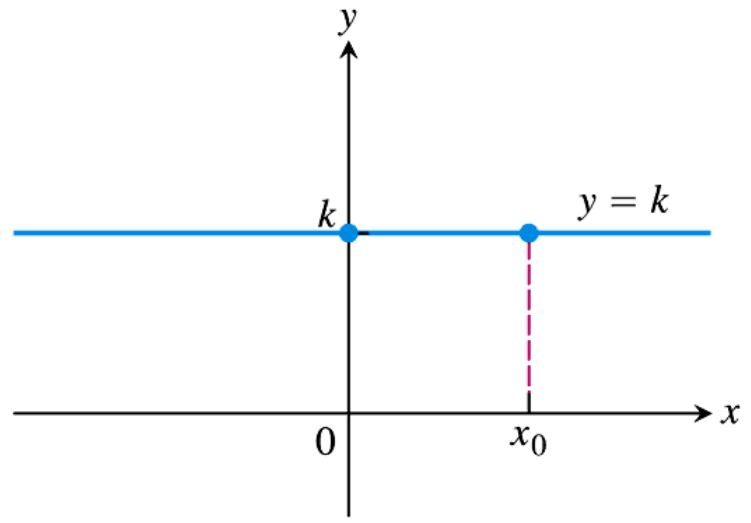


(a) Identity function

Identity and Constant Functions

- If f is the **constant function** $f(x)=k$ (function with the constant value k), then for any value of x_0,

$$\lim_{x \to x_0} f(x) = \lim_{x \to x_0} k = k$$



(b) Constant function

THEOREM 1 Limit Laws

If L, M, c and k are real numbers and

$$\lim_{x \to c} f(x) = L \quad \text{and} \quad \lim_{x \to c} g(x) = M, \quad \text{then}$$

1. *Sum Rule*: $\lim_{x \to c} (f(x) + g(x)) = L + M$

The limit of the sum of two functions is the sum of their limits.

2. *Difference Rule*: $\lim_{x \to c} (f(x) - g(x)) = L - M$

The limit of the difference of two functions is the difference of their limits.

3. *Product Rule*: $\lim_{x \to c} (f(x) \cdot g(x)) = L \cdot M$

The limit of a product of two functions is the product of their limits.

4. *Constant Multiple Rule:* $\lim_{x \to c} (k \cdot f(x)) = k \cdot L$

The limit of a constant times a function is the constant times the limit of the function.

5. *Quotient Rule*: $\lim_{x \to c} \dfrac{f(x)}{g(x)} = \dfrac{L}{M}, \quad M \neq 0$

The limit of a quotient of two functions is the quotient of their limits, provided the limit of the denominator is not zero.

6. *Power Rule*: If r and s are integers with no common factor and $s \neq 0$, then

$$\lim_{x \to c} (f(x))^{r/s} = L^{r/s}$$

provided that $L^{r/s}$ is a real number. (If s is even, we assume that $L > 0$.)

The limit of a rational power of a function is that power of the limit of the function, provided the latter is a real number.

Examples

Find the limit of (a) $\lim_{x \to c}\left(x^3 + 4x^2 - 3\right)$ (b) $\lim_{x \to c}\frac{\left(x^4 + x^2 - 1\right)}{x^2 + 5}$

Ans:

$$\text{(a)}\ \lim_{x \to c}\left(x^3 + 4x^2 - 3\right) = \lim_{x \to c} x^3 + \lim_{x \to c} 4x^2 - \lim_{x \to c} 3$$
$$= c^3 + 4c^2 - 3$$

$$\text{(b)}\ \lim_{x \to c}\frac{\left(x^4 + x^2 - 1\right)}{x^2 + 5} = \frac{\lim_{x \to c}\left(x^4 + x^2 - 1\right)}{\lim_{x \to c}\left(x^2 + 5\right)}$$
$$= \frac{\lim_{x \to c} x^4 + \lim_{x \to c} x^2 - \lim_{x \to c} 1}{\lim_{x \to c} x^2 + \lim_{x \to c} 5} = \frac{\left(c^4 + c^2 - 1\right)}{c^2 + 5}$$

Using the Product Rule, find the limit of $\lim_{x\to 0}\frac{\tan x}{x}$

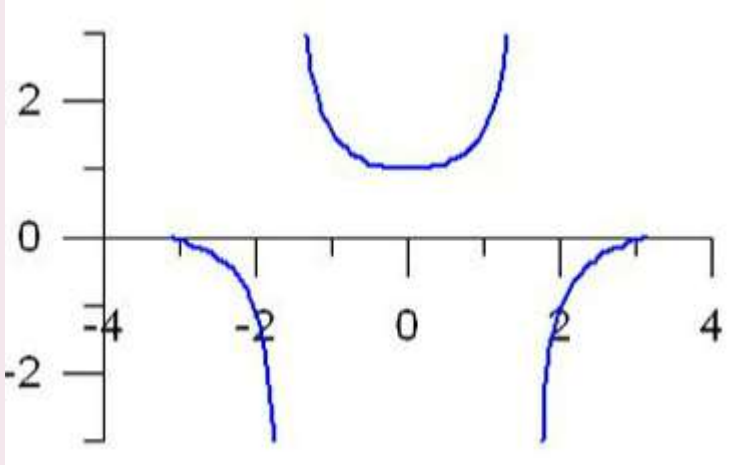
Ans:

$$\lim_{x\to 0}\frac{\tan x}{x}=\lim_{x\to 0}\frac{\sin x}{x}\frac{1}{\cos x}$$

$$=\lim_{x\to 0}\frac{\sin x}{x}\times\lim_{x\to 0}\frac{1}{\cos x}$$

$$=1\times\frac{1}{\cos 0}=1$$

Lecture 4



THEOREM 2—Limits of Polynomials

If $P(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_0$, then

$$\lim_{x \to c} P(x) = P(c) = a_n c^n + a_{n-1}c^{n-1} + \cdots + a_0.$$

THEOREM 3—Limits of Rational Functions

If $P(x)$ and $Q(x)$ are polynomials and $Q(c) \neq 0$, then

$$\lim_{x \to c} \frac{P(x)}{Q(x)} = \frac{P(c)}{Q(c)}.$$

Examples

Find the limit of (a) $\lim_{x\to 3}\left(x^2(2-x)\right)$ (b) $\lim_{x\to 2}\frac{\left(x^2+2x+4\right)}{x+2}$

Ans:

$$\text{(a)}\ \lim_{x\to 3}\left(x^2(2-x)\right)=3^2(2-3)=-9$$

$$\text{(b)}\ \lim_{x\to 2}\frac{\left(x^2+2x+4\right)}{x+2}=\frac{\left(2^2+2(2)+4\right)}{2+2}=3$$

Eliminating Zero Denominators Algebraically

Example

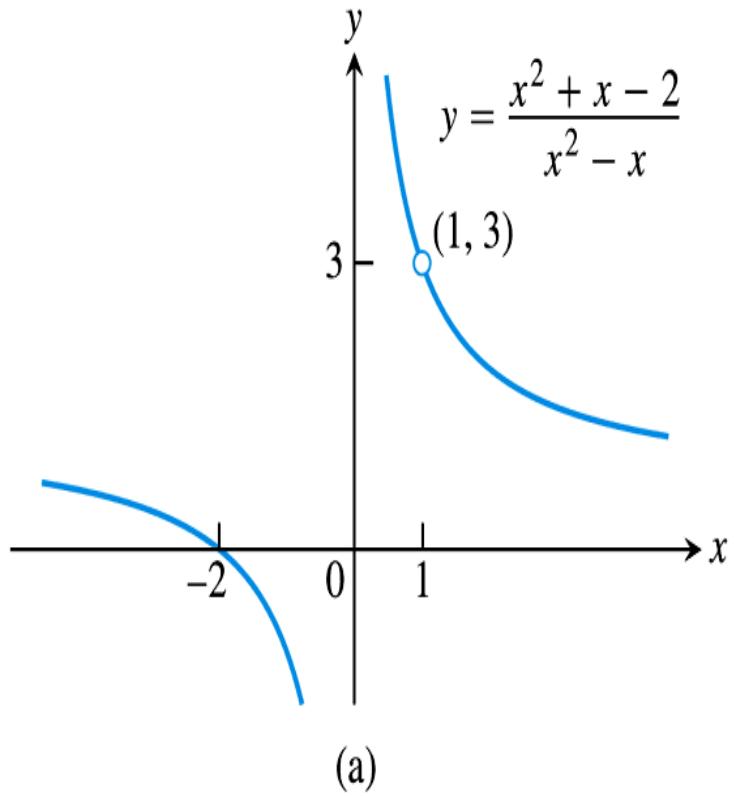
Evaluate $\lim_{x \to 1} \frac{x^2 + x - 2}{x^2 - x}$

$$\lim_{x \to 1} \frac{x^2 + x - 2}{x^2 - x}$$

$$= \lim_{x \to 1} \frac{(x+2)(x-1)}{x(x-1)}$$

$$= \lim_{x \to 1} \frac{x+2}{x}$$

$x \neq 0$



(a)

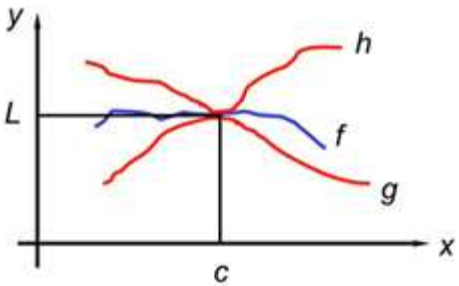
The Sandwich Theorem

THEOREM 4 The Sandwich Theorem

Suppose that $g(x) \leq f(x) \leq h(x)$ for all x in some open interval containing c, except possibly at $x = c$ itself. Suppose also that

$$\lim_{x \to c} g(x) = \lim_{x \to c} h(x) = L.$$

Then $\lim_{x \to c} f(x) = L$.

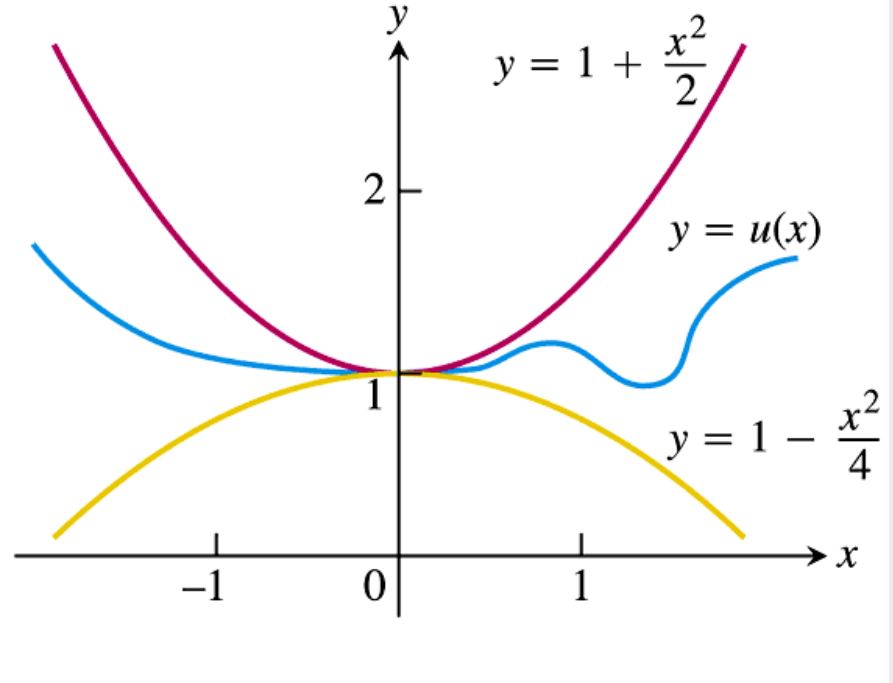


Example 1

Given that

$$1-\frac{x^2}{4} \le u(x) \le 1+\frac{x^2}{2}$$

for all $x \neq 0$, find $\lim_{x\to 0} u(x)$



$$\lim_{x\to 0} u(x) = 1$$

Example 2

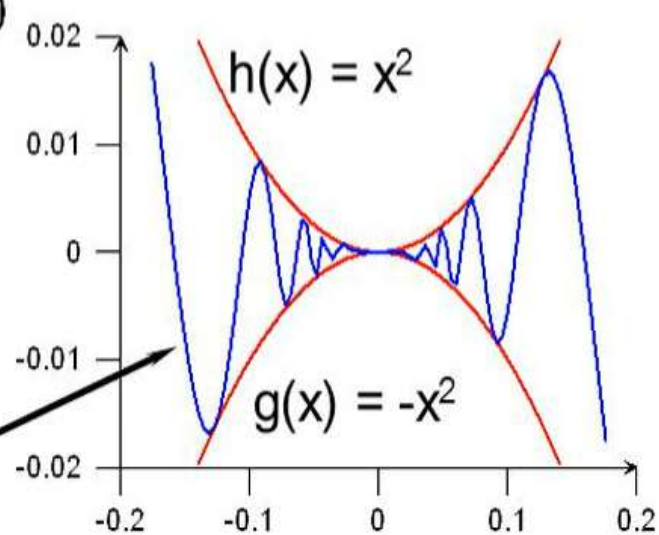
Show that $\lim_{x\to 0} x^2 \sin(1/x) = 0$

Ans:

since $-x^2 \le x^2 \sin(1/x) \le x^2$

and $\lim_{x\to 0}(-x^2) = \lim_{x\to 0}(x^2) = 0$

Finally, $\lim_{x\to 0} x^2 \sin(1/x) = 0$



Example 3

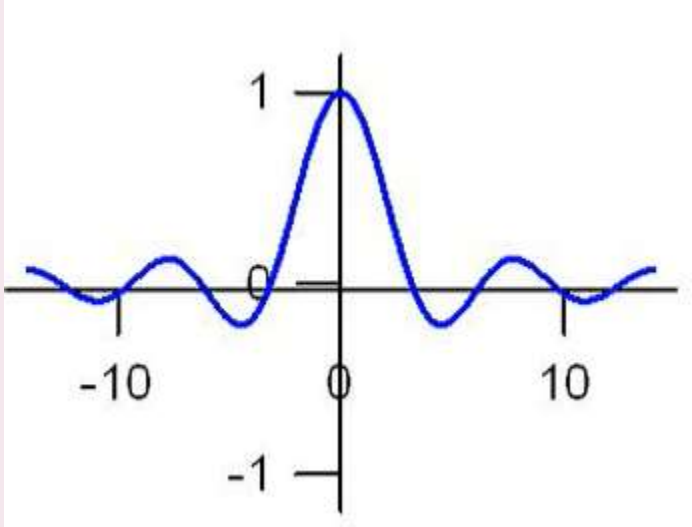
Find $\lim_{x\to\infty} \frac{\sin x}{x}$

since, $-1 \le \sin x \le 1$, for $x > 0$

$$\frac{-1}{x} \le \frac{\sin x}{x} \le \frac{1}{x}$$

by Sandwich theorem

$$0 = \lim_{x\to\infty}\left(\frac{-1}{x}\right) = \lim_{x\to\infty}\frac{\sin x}{x} = \lim_{x\to\infty}\frac{1}{x}$$



The limit of $f(x)$ as x -> a equals the number L

- Need to show that
  - The gap between $f(x)$ and L can be made "as small as we choose" if x is kept "close enough" to a.
  - What happens if we specify the size of this gap between $f(x)$ and L

Example 1

- Consider the function $y = 2x - 1$ near $a = 4$.
  - We know, $\lim_{x \to 4} (2x - 1) = 7$
- How close to $a = 4$ does x have to be so that $y = 2x - 1$ differs from 7 by, say, less than 2 units?
  - i.e. for what values of x is $|y - 7| < 2$?
    - Express $|y - 7|$ in terms of x
    - Solve the inequality

- Solution: we are asked: for what values of *x is* $|y - 7|<2$?

Steps: $|y - 7|=|(2x-1)-7|=|2x-8|$

- What values of x satisfy the inequality $|2x-8|<2$?

Steps : $|2x-8|<2$

$-2<2x-8<2$

$6<2x<10$

$3<x<5$

$-1<x-4<1$

Thus, keeping x within 1 unit of $x_0 = 4$ will keep *y* within 2 units of $y_0 = 7$.

Example 1 (cont.)

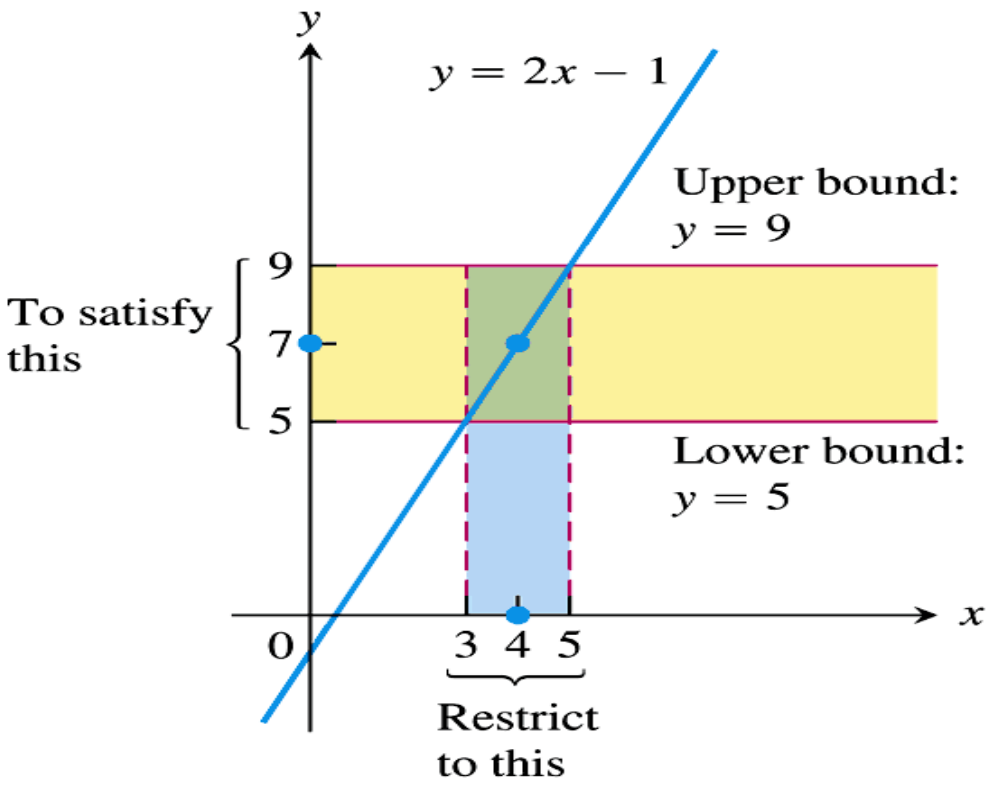


FIGURE 2.15 Keeping x within 1 unit of $x_0 = 4$ will keep y within 2 units of $y_0 = 7$ (Example 1).

What have you learnt?

- LIMIT
  - Informal definition
  - Precise definition
  - Find limit of functions
  - Use some limit laws
  - Exposed to theorems involving limits